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Visual feedback for video muted participants in an online meeting

Abstract

One or more animated files are generated using an image of a user and stored in a data store associated with the user. It is determined that the user is participating in an online meeting and the user is not transmitting video. The one or more animated files are obtained from the data store and transmitted for display playback on a loop on devices of participants participating in the online meeting with the user when the user is not transmitting video.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to online video meetings/conferences.

BACKGROUND

(2) Although audio may be considered the most important media in multimedia conference calls,

visual feedback is also important. However, some people are not always comfortable sharing video, especially in medium or large conferences or when a person is not directly contributing to a conversation. If a user has uploaded a picture, other participants may get a feeling of presence and an idea of who the user is. However, a static photograph of the user can be non-informative and does not give any feedback of the user's reactions during the online meeting.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a block diagram of an online video conference/meeting system configured to support providing visual feedback for video muted participants, according to an example embodiment.
- (2) FIG. 2 illustrates an example user interface in which visual feedback for video muted participants is provided, according to an example embodiment.
- (3) FIG. 3 is a flow diagram illustrating a method of providing visual feedback for video muted participants, according to an example embodiment.
- (4) FIG. 4 is a flow diagram illustrating another method of providing visual feedback for video muted participants, according to an example embodiment.
- (5) FIG. 5 is a hardware block diagram of a device that may be configured to perform the endpoint-based operations involved in providing visual feedback for video muted participants, according to an example embodiment.
- (6) FIG. 6 is a hardware diagram of a computer device that may be configured to perform the meeting server operations involved in providing visual feedback for video muted participants, according to an example embodiment.

DETAILED DESCRIPTION

Overview

(7) In one embodiment, a computer-implemented method is provided for improving a visual feedback of video muted participants in an online meeting or communication session. The method includes generating one or more animated files using an image of a user and storing the one or more animated files in a data store associated with the user. It is determined that a user is participating in an online meeting and the user is not transmitting video. The one or more animated files are obtained from the data store and transmitted for display playback on a loop on devices of participants participating in the online meeting with the user when the user is not transmitting video.

Example Embodiments

(8) Artificial Intelligence (AI) and machine learning (ML) technologies have been used to create animations of still pictures in which the still pictures appear to be transformed into “live videos.” Using AI and ML technologies, photos taken from any camera may be “brought to life” by using pre-recorded driver videos of facial movements and applying the pre-recorded driver video that works the best for a particular still photo. The processed still photos appear to show people in action by animating the people so that the people, for example, move their heads, blink their eyes, smile, laugh, etc.

(9) Embodiments described herein provide for animating pictures of people during an online meeting or communication session for improved video feedback. Embodiments described herein create a more dynamic scene in a conference setting by animating a user's current profile photo. In particular, one or more basic animated files may be created using the user's profile photo by animating the profile photo so that, for example, the user's head moves, the user's eyes blink, the user smiles, etc. In addition, several emotion animated files may be created using the user's profile photo. Each emotion animated file may be associated with a different emotion or reaction. For example, a nodding animated file may be created from the user's profile photo in which the user's

head nods. As another example, a laughing animated file may be created from the user's profile photo in which the user appears to laugh.

(10) In one embodiment, the one or more basic animated files may be played on a loop when the user is not transmitting video to create an illusion that the picture of the user is a real person and not merely a still image. In another embodiment, an emotion animated file may be played in response to obtaining a trigger received from a user. The trigger may be obtained, for example, using face or gesture detection or from receiving an input from a user (e.g., when the user selects a reaction button). The animated emotion files may provide an indication that the user is engaged in and reacting to the online communication session.

(11) In addition, embodiments described herein provide conference participants with a more vibrant grid view. For example, instead of viewing a grid of still pictures or initials of users' names, the grid view may show photos of participants with slight movement/animation to the photos. Instead of mimicking every move or expression of the participants, embodiments described herein create an illusion that a user is viewing a group of real people and not just still images. In addition, a user may be able to detect visual feedback from other participants when the participants trigger animated emotion files to, for example, show approval (e.g., by nodding) or amusement (e.g., by smiling/laughing).

(12) Reference is first made to FIG. 1. FIG. 1 shows a block diagram of a system **100** that is configured to provide improved visual feedback of video muted participants in an online meeting. The system **100** includes one or more meeting server(s) **110** and a plurality of user devices **120-1** to **120-N** (sometimes referred to herein individually as user device **120** or collectively as user devices **120**) that communicate with meeting server(s) **110** and cloud **140** via one or more networks **130**.

(13) The meeting server(s) **110** are configured to provide an online meeting service for hosting a communication session among user devices **120-1** to **120-N**. In some embodiments, the meeting server(s) **110** may generate animated files based on images/pictures/photographs of users. As described further below, the meeting server(s) **110** may additionally transmit different animated files to participants in an online meeting based on triggers received from user devices **120-1** to **120-N**.

(14) The user devices **120-1** to **120-N** may be tablets, laptop computers, desktop computers, Smartphones, virtual desktop clients, or any user device now known or hereinafter developed that can run a meeting client or similar communication functionality. The user devices **120-1** to **120-N** may have a dedicated physical keyboard or touch-screen capabilities to provide a virtual on-screen keyboard to enter text. The network(s) **130** may include wired local and wide-area networks as well as wireless local and wide-area networks. The user devices **120-1** to **120-N** may also have short-range wireless system connectivity (such as Bluetooth™ wireless system capability, ultrasound communication capability, etc.) to enable local wireless connectivity with a videoconference endpoint in a meeting room or with other user devices in the same meeting room. In some embodiments, user devices **120-1** to **120-N** may be videoconference endpoints, such as a personal videoconference endpoint.

(15) Cloud **140** may include a memory **122** that includes a file store **124**. File store **124** may store animated files associated with different users/participants, such as users of user devices **120-1** to **120-N**. File store **124** may store basic files **126-1** to **126-N** (sometimes referred to herein individually as basic file **126** and collectively as basic files **126**) and emotion files **128-1** to **128-N** (sometimes referred to herein individually as emotion file **128** and collectively as emotion files **128**). In some embodiments, each user/participant may be associated with one or more basic files **126**. In other embodiments, each user/participant may additionally be associated with a plurality of emotion files **128**.

(16) Meeting server(s) **110** or another device may create one or more animated files for a user based on the user's profile picture using one or more ML models and/or AI technologies. The user's profile picture may be, for example, a photograph of the user, a cartoon avatar of the user, a

picture/drawing that is not of the user (e.g., a cat, a dog, etc.), etc. A user may upload the profile picture for a user profile, a profile picture may automatically be uploaded from a company directory, or the meeting system may obtain the user profile from other sources (e.g., social media services). The animated files created from the profile pictures may be GIFs or other types of files that are video representations of the users based on the users' profile pictures. Each animated file may be short in duration (e.g., a few seconds long). A basic animated file (such as one of the basic files **126-1** to **126-N**) may be created based on the profile picture in which the user appears to make subtle movements in a neutral manner. For example, when the profile picture is a photograph or image of the user, the user's head may move slightly, the user's eyes may blink, etc., while the user maintains a neutral expression. In some embodiments, more than one basic animated file may be created for each user. In this embodiment, each basic animated file may show the user making different subtle movements. Other types of profile pictures (e.g., a cartoon, an animal) may be animated in a similar manner.

(17) The basic file(s) **126** may be created using an AI model and in a way that is a natural loop, which limits size and computational load. The user may have a common expression at the beginning and the end of the basic file **126**, so that the looping of the basic file **126** is smooth and relatively undetectable. In addition, when more than one basic file **126** is created for a user, each basic file **126** has the same common expression at the beginning and end so that switching between basic files **126** is smooth and seamless. In this embodiment, the basic files **126** may be played in random order to break any repetitive appearance. Each basic file **126** may be associated with a user identifier corresponding to the user and a file identifier associated with the basic file **126**.

(18) In addition to the basic file(s) **126**, one or more emotion animated files (such as emotion files **128-1** to **128-N**) may be generated using the user's profile picture. Each emotion file **128** may be associated with a different reaction or emotion. For example, one emotion file **128** may be associated with agreement and may show the user nodding in agreement. Another emotion file **128** may be associated with amusement and may show the user smiling or laughing. Each emotion file **128** may be relatively short and the user may have a common start/stop expression in each emotion file. The common start/stop expression may be the same as the common start/stop expression in the basic file(s) **126**, which makes for seamless transitions between emotion files **128** or between an emotion file **128** and a basic file **126**. Each emotion file **128** may be associated with a user identifier indicating the user depicted in the file and an emotion identifier associated with the emotion or reaction corresponding to the emotion file **128**.

(19) In one embodiment, during a videoconference/multimedia session in which a user is not transmitting video, meeting server(s) **110** may obtain the animated files from file store **124** and transmit the animated files as the video of the user. In this embodiment, meeting server(s) **110** may download the animated files for one or more participants of an online meeting from cloud **140** at the beginning of the online meeting. The meeting server(s) **110** may determine that one or more of the participants wish to transmit their basic file(s) **126** and, optionally, one or more emotion files **128** in place of transmitting video (e.g., when the participants have their videos muted, based on an input from the participants, etc.) and the meeting server(s) **110** may transmit the animated files to the other participants during the meeting. In some embodiments, the meeting server(s) may use the animated files as a source (e.g., instead of camera images), which is encoded and sent to the participants of the online meeting as a video stream. In other embodiments, the animated files may be transmitted to the participants.

(20) In another embodiment, the animated files may be stored locally on each participant's device/client and the device/client may transmit the animated files as the video of the user. For example, each user device **120-1** to **120-N** may obtain the animated files for the user of the user device **120** from cloud **140** at the beginning of an online meeting/videoconference (e.g., automatically at the beginning of each online meeting/videoconference or in response to a selection by a user, such as a user selecting a video mute option). In other embodiments, user devices **120-1**

to **120-N** may store the animated files and the user device **120** does not download the animated files from cloud **140** at the beginning of each videoconference. In some embodiments, user devices **120-1** to **120-N** may have integrated animation generators and the user devices **120-1** to **120-N** may generate the animated files for the user. In these embodiments, the animation generators may access third party AI/ML models to generate the animated files and upload the animated files to user devices **120-1** to **120-N**.

(21) When the animated files are stored at a user device **120**, the user device **120** may determine that the user wishes to transmit the animated files instead of a video and the user device **120** may transmit basic file(s) **126** and, optionally, one or more of the emotion files **128** during the online meeting. The user device **120** may use the animated files as a source that is encoded and transmitted to the participants of the online meeting as a video stream or the user device **120** may transmit the animated files to the participants. In some embodiments, the user device **120** may determine that the animated files are to be transmitted as the video of the user when the user's video is turned off/muted or the user has closed the camera lens cover. In other embodiments, the user device **120** may determine that the animated files are to be transmitted as video based on an indication or selection from the user.

(22) In other embodiments, for small/medium teleconferences, each participating user device **120-1** to **120-N** may fetch basic files **126-1** to **126-N** and emotion files **128-1** to **128-N** for other participants in the teleconference from cloud **140** and render the animated files locally. In this situation, a user device **120** may receive an indication (e.g., from meeting server(s) **110** or other user devices **120-1** to **120-N**) that one or more participants wish to display the animated files instead of video and user device **120** may display the animated files for those participants during the teleconference.

(23) In some embodiments, the user device **120** or the meeting server(s) may obtain only the basic file(s) **126** for a user. For example, a user may select an operating mode in which only the basic file(s) **126** are being transmitted and emotional files **128** are not being transmitted. In this example, if a user subsequently selects an operating mode in which the emotion files **128** are to be used, the user device **120** or the meeting server(s) may fetch and store the emotion files **128**.

(24) When a videoconference/multimedia session begins and a user has indicated that the user's animated files are to be used as a source/transmitted instead of video (e.g., by muting video, selecting an option to transmit the animated files, etc.), the user's basic file(s) **126** may be played on a loop to other participants in the meeting. As described above, when the user is associated with more than one basic file **126**, the basic files **126** may be played on a loop (e.g., in a random order) in place of video. When a user selects the operating mode in which the emotion files(s) **128** are to be used and a user reacts or displays an emotion, a user device **120-1** to **120-N** or meeting server(s) **110** may receive a trigger indicating that a corresponding emotion file **128** is to be transmitted/displayed.

(25) In some embodiments, the emotion/reaction may be triggered based on face detection and/or gesture detection. For example, a camera on the user's user device **120** may detect that the user is finding something amusing/funny by smiling or laughing or that the user is nodding in agreement to something that was said during the teleconference. Although the camera is not transmitting video of the user to meeting server(s) **110**, the camera may still capture video of the user locally and may detect the user's gestures. The user may select an option in which the gesture/facial recognition is turned off (e.g., by selecting a mode in which the emotion files **128** are not transmitted, selecting an option to trigger an emotion file **128** using a reaction button, etc.). In other embodiments, the emotion/reaction may be triggered based on receiving an input from the user. For example, as discussed below with respect to FIG. 2, a user may select a reaction button from a videoconference interface to trigger the emotion/reaction.

(26) When the emotion/reaction is triggered by the user, the user's user device **120** (or a videoconference client/application on the user device **120**) transmits a signal to meeting server(s)

110 (e.g., when the animated files are stored at and transmitted by meeting server(s) **110**) or to a local client/software (when the animated files are stored at and transmitted by user device **120**) indicating that the emotion/reaction has been triggered. The signal includes a participant identifier (ID) associated with the user and an emotion ID associated with the reaction/emotion and the corresponding emotion file **128**. When the signal is received (e.g., at meeting server(s) **110** or the local client/software at user device **120**), the emotion file **128** corresponding to the user ID and the emotion ID is transmitted to the other participants in the teleconference for display playback on their user devices **120**. In some embodiments, the emotion file **128** may be used as a source/transmitted at the conclusion of a current loop of a basic file **126**. Since the basic file **126** and the emotion file **128** include a common start/stop expression for the user, using or transmitting the emotion file **128** when the basic file **126** concludes may provide a seamless transition between files. In other embodiments, the emotion file **128** may be used as a source or transmitted before a conclusion of the basic file **126**. In this embodiment, the switch between the basic file **126** and the emotion file **128** may be more glitchy or unnatural, but the emotion/expression may be conveyed in a timelier manner. At the conclusion one or more loops of the emotion file **128**, the basic file(s) **126** may be once again played on a loop until an additional emotion trigger is received for the user.

(27) Reference is now made to FIG. 2 with continued reference to FIG. 1. FIG. 2 illustrates an example user interface **200** in which animated files may be received as video for one or more participants in a videoconference/multimedia session. User interface **200** includes a grid view of feeds **202-1** to **202-9** associated with different participants.

(28) As illustrated in FIG. 2, participants associated with feeds **202-1**, **202-2**, and **202-6** are displaying video and participants associated with feeds **202-3** and **202-9** are displaying initials with no video and no profile picture. Participants associated with feeds **202-4**, **202-5**, **202-7**, and **202-8** are displaying profile pictures instead of transmitting video. In addition, feeds **202-4** and **202-5** include video muted indicators **204-1** and **204-2**. Video muted indicators **204-1** and **204-2** may optionally be displayed when a user is transmitting animated files in the place of video. Since the animated files should not look like or be confused with a regular video participant, video muted indicators **204-1** and **204-2** may be displayed to ensure that the user is aware that the participants associated with feeds **202-4** and **202-5** are not transmitting live video. Feed **202-7** does not include a video muted indicator and, in this example, the user associated with feed **202-7** is displaying a picture and not using or transmitting animated files.

(29) As illustrated in FIG. 2, the feeds **202-4** and **202-5** that are using animated files look like feed **202-6** that is displaying an image of the user except that the images in feeds **202-4** and **202-5** are animated instead of static. In practice, the profile pictures associated with feeds **202-4** and **202-5** would appear to make slight gestures and, optionally, react during the online meeting.

(30) User interface **200** may additionally include button **206** to start or stop transmitting video. As illustrated in FIG. 2, the button **206** says "Start video," which indicates that the user's video is turned off. In some embodiments, a user may make a selection to indicate that animated files are to be used as a source/transmitted in place of video by default when the user's video is turned off or muted. In this embodiment, when button **206** indicates that the video is turned off, it will indicate to the user that the animated files are being used as a source/transmitted. To turn the video on and stop using or transmitting the animated files, the user may select button **206**. In other embodiments, the user may select whether to use or transmit animated files when the video is turned off. In this embodiment, button **208**, which indicates whether the animated files are being used or transmitted, may be displayed to the user. In the example, shown in FIG. 2, button **208** says "Stop animation," which indicates that the animated files are being used as a source/transmitted. To stop using or transmitting the animated files, the user may select button **208**. When the user's video is off and the user has chosen to stop the animation, the user's profile picture, initials, or something else may be displayed in place of the animation. When the user's video is being transmitted, a user may not be able to select button **208**. In some embodiments, user interface **200** may display buttons with

additional options, such as an option to transmit only basic file(s) **126** (and not emotion files **128**), an option to not allow the camera to detect emotions, etc.

(31) In the example illustrated in FIG. **2**, the user's video is turned off/muted. When a user wishes to trigger an emotion file **128** (e.g., when the user has selected an option to not allow the camera to perform facial/gesture recognition), the user may select button **210**, which displays menu **212**. Menu **212** displays graphics that represent different emotions/reactions. A user may select one of the graphics to trigger an emotion file **128** to be transmitted and displayed for playback. For example, the user may select the laughing face graphic to indicate that something is funny or amusing and an emotion file **128** showing the user laughing may be transmitted. In some embodiments, a user may provide emotions/reactions using menu **212** in addition to triggering emotions/reactions using face and gesture detection.

(32) In some embodiments, triggering the emotion/reaction using menu **212** may draw more attention to the emotion/reaction than triggering the emotion/reaction using face or gesture recognition. For example, emotions/reactions selected from menu **212** may be associated with separate emotion files **128** for extreme or funny animations of the user laughing, clapping, wowing, etc. In some embodiments, prerecorded audio or stock audio may be played with the extreme or funny animation files. Use of the extreme or funny emotion animated files may be activated and deactivated by the user or a host of the videoconference.

(33) By displaying animated files in place of profile pictures, even in the most basic mode, the grid view is more dynamic and provides a user with a feeling of participants being present in an online meeting or videoconference. In addition, by giving the users an option to provide emotions/reactions, a user may participate in a meeting nonverbally (e.g., by triggering agreement or amusement emotions/reactions) without displaying video. The emotion animated files may additionally give a speaker feedback regarding participants' emotions and reactions when the participants' audio and video are muted.

(34) Referring to FIG. **3**, FIG. **3** is a flow chart of a method **300** for providing visual feedback for video muted participants of an online meeting/multimedia session, according to embodiments described herein. Method **300** may be performed by meeting server(s) **110** and/or user devices **120-1** to **120-N**.

(35) At **302**, an image of a user may be uploaded to a user database. For example, a user or another person (e.g., an administrator) may upload an image/photograph as a profile picture or the image/photograph may be automatically uploaded from a directory or another source. At **304**, one or more machine learning models may be applied to create at least one general/basic animated file and optionally several emotional animated files based on the image of the user. For example, ML models and/or AI technologies may be applied to the image/photograph of the user to create basic file(s) **126** in which the user moves slightly in a neutral manner. Optionally, emotion files **128** may additionally be created in which the user appears to perform gestures associated with different emotions/reactions (e.g., smiling, laughing, nodding, etc.). The basic file(s) **126** and optionally the emotion files **128** are stored to a user database, such as file store **124**, a file store at user device **120**, or another database.

(36) At **306**, it is detected that a first user is not transmitting video during an online meeting/multimedia session. For example, a user device **120** or meeting server(s) **110** may detect that a user is not transmitting video during an online meeting/multimedia session. The user may have selected an option to “mute” video, such as by pressing button **206** in FIG. **2**.

(37) At **308**, at least the general/basic animated files are fetched from a user database and displayed in the display area. For example, user device **120** and/or meeting server(s) may obtain basic file(s) **126** for the user and transmit the basic file(s) **126** on a loop for display playback on user devices **120** of other participants in the online meeting/multimedia session. If one basic file **126** is obtained, the basic file **126** may be played on a loop. If more than one basic file **126** is obtained, the basic files **126** may be played on a loop (e.g., in a random order) to break any repetitive appearance.

(38) At **310**, the user device may monitor facial features/gestures of the local user to detect one of a predefined set of emotions when emotions are selected as the operating mode. For example, a user may select an option to allow emotion files **128** to be transmitted and to allow facial/gesture detection to be used. A camera of user device **120** may monitor the user's facial features/gestures and detect a particular predefined emotion/reaction (e.g., laughing, nodding, etc.) that corresponds to an emotion/gesture associated with an emotion file **128**.

(39) At **312**, the user device may send a control signal to the meeting server identifying the user and the emotion when the predefined emotion is detected. For example, when the meeting server(s) **110** has obtained the basic file(s) **126** and the emotion files **128**, user device **120** may transmit a control signal to meeting server(s) **110** when the predefined emotion has been detected. The control signal may include a user ID associated with a user and an emotion ID associated with the detected emotion.

(40) At **314**, the emotion animated file corresponding to the identified user and emotion is played in response to receiving the control signal. For example, meeting server(s) **110** may obtain the emotion file **128** corresponding to the user and the detected emotion and meeting server(s) **110** may transmit the emotion file **128** for display playback on the user devices **120** of other participants in the online meeting/multimedia session. In this way, the other users may receive visual feedback from the user when the user's camera is not turned on.

(41) At **316**, the general animation file is once again transmitted after looping or until a new control signal is received for the user. For example, after the emotion file **128** is played one or more times on a loop, the basic file(s) **126** are once again transmitted for the user. The basic file(s) may continue to be played on a loop until a new control signal is received for the user (e.g., in response to detecting an emotion/reaction corresponding to an emotion file **128**, receiving an input associated with an emotion file **128**, etc.).

(42) Referring to FIG. 4, FIG. 4 is a flow diagram of a method **400** for providing visual feedback for video muted participants of an online meeting/multimedia session, according to embodiments described herein. Method **400** may be performed by meeting server(s) **110** and/or user devices **120-1** to **120-N**.

(43) At **410**, one or more animated files are generated using an image of a user. For example, one or more ML models may be applied to an image/photograph of a user to generate one or more animated files. At **420**, the one or more animated files are stored in a data store associated with the user. For example, the one or more animated files may be stored at a cloud, such as in file store **124** at cloud **140** of FIG. 1, at a data store at the user's user device (such as user device **120-1**), or in a different location.

(44) At **430**, it is determined that the user is participating in an online meeting and the user is not transmitting video. For example, the user may be participating in an online meeting with other participants and the user may have made a selection indicating that the video is “muted” or that animated files are to be used in place of video. User device **120-1** or meeting server(s) **110** may determine that the user is not transmitting video based on the user's selection.

(45) At **440**, one or more animated files may be obtained from the data store. For example, meeting server(s), user device **120-1**, or another device may obtain the one or more animated files associated with the user from the data store. At **450**, the one or more animated files are transmitted for display playback on a loop on devices of participants participating in the online meeting with the user when the user is not transmitting video. For example, meeting server(s) **110** or user device **120-1** may transmit the one or more animated files for display playback on a loop on user devices **120-2** to **120-N**.

(46) Referring to FIG. 5, FIG. 5 illustrates a hardware block diagram of a computing/computer device **500** that may perform functions of a user device associated with operations discussed herein in connection with the techniques depicted in FIGS. 1-4. In various embodiments, a computing device, such as computing device **500** or any combination of computing devices **500**, may be

configured as any devices as discussed for the techniques depicted in connection with FIGS. 1-4 in order to perform operations of the various techniques discussed herein.

(47) In at least one embodiment, the computing device **500** may include one or more processor(s) **502**, one or more memory element(s) **504**, storage **506**, a bus **508**, one or more network processor unit(s) **510** interconnected with one or more network input/output (I/O) interface(s) **512**, one or more I/O interface(s) **514**, and control logic **520**. In various embodiments, instructions associated with logic for computing device **500** can overlap in any manner and are not limited to the specific allocation of instructions and/or operations described herein.

(48) In at least one embodiment, processor(s) **502** is/are at least one hardware processor configured to execute various tasks, operations and/or functions for computing device **500** as described herein according to software and/or instructions configured for computing device **500**. Processor(s) **502** (e.g., a hardware processor) can execute any type of instructions associated with data to achieve the operations detailed herein. In one example, processor(s) **502** can transform an element or an article (e.g., data, information) from one state or thing to another state or thing. Any of potential processing elements, microprocessors, digital signal processor, baseband signal processor, modem, PHY, controllers, systems, managers, logic, and/or machines described herein can be construed as being encompassed within the broad term 'processor'.

(49) In at least one embodiment, memory element(s) **504** and/or storage **506** is/are configured to store data, information, software, and/or instructions associated with computing device **500**, and/or logic configured for memory element(s) **504** and/or storage **506**. For example, any logic described herein (e.g., control logic **520**) can, in various embodiments, be stored for computing device **500** using any combination of memory element(s) **504** and/or storage **506**. Note that in some embodiments, storage **506** can be consolidated with memory element(s) **504** (or vice versa), or can overlap/exist in any other suitable manner.

(50) In at least one embodiment, bus **508** can be configured as an interface that enables one or more elements of computing device **500** to communicate in order to exchange information and/or data. Bus **508** can be implemented with any architecture designed for passing control, data and/or information between processors, memory elements/storage, peripheral devices, and/or any other hardware and/or software components that may be configured for computing device **500**. In at least one embodiment, bus **508** may be implemented as a fast kernel-hosted interconnect, potentially using shared memory between processes (e.g., logic), which can enable efficient communication paths between the processes.

(51) In various embodiments, network processor unit(s) **510** may enable communication between computing device **500** and other systems, entities, etc., via network I/O interface(s) **512** (wired and/or wireless) to facilitate operations discussed for various embodiments described herein. Examples of wireless communication capabilities include short-range wireless communication (e.g., Bluetooth), wide area wireless communication (e.g., 4G, 5G, etc.). In various embodiments, network processor unit(s) **510** can be configured as a combination of hardware and/or software, such as one or more Ethernet driver(s) and/or controller(s) or interface cards, Fibre Channel (e.g., optical) driver(s) and/or controller(s), wireless receivers/transmitters/transceivers, baseband processor(s)/modem(s), and/or other similar network interface driver(s) and/or controller(s) now known or hereafter developed to enable communications between computing device **500** and other systems, entities, etc. to facilitate operations for various embodiments described herein. In various embodiments, network I/O interface(s) **512** can be configured as one or more Ethernet port(s), Fibre Channel ports, any other I/O port(s), and/or antenna(s)/antenna array(s) now known or hereafter developed. Thus, the network processor unit(s) **510** and/or network I/O interface(s) **512** may include suitable interfaces for receiving, transmitting, and/or otherwise communicating data and/or information in a network environment.

(52) I/O interface(s) **514** allow for input and output of data and/or information with other entities that may be connected to computer device **500**. For example, I/O interface(s) **514** may provide a

connection to external devices such as a keyboard **525**, keypad, a touch screen, and/or any other suitable input and/or output device now known or hereafter developed. This may be the case, in particular, when the computer device **500** serves as a user device described herein. In some instances, external devices can also include portable computer readable (non-transitory) storage media such as database systems, thumb drives, portable optical or magnetic disks, and memory cards. In still some instances, external devices can be a mechanism to display data to a user, such as, for example, a computer monitor, a display screen, such as display **530** shown in FIG. 5, particularly when the computer device **500** serves as a user device as described herein. Display **530** may have touch-screen display capabilities. Additional external devices may include a video camera **535** and microphone/speaker combination **540**. In addition, the computing device **500** may further include an ultrasound transceiver **545** that may be used for the various purposes described herein.

(53) In various embodiments, control logic **520** can include instructions that, when executed, cause processor(s) **502** to perform operations, which can include, but not be limited to, providing overall control operations of computing device; interacting with other entities, systems, etc. described herein; maintaining and/or interacting with stored data, information, parameters, etc. (e.g., memory element(s), storage, data structures, databases, tables, etc.); combinations thereof, and/or the like to facilitate various operations for embodiments described herein.

(54) The programs described herein (e.g., control logic **520**) may be identified based upon application(s) for which they are implemented in a specific embodiment. However, it should be appreciated that any particular program nomenclature herein is used merely for convenience; thus, embodiments herein should not be limited to use(s) solely described in any specific application(s) identified and/or implied by such nomenclature.

(55) In various embodiments, entities as described herein may store data/information in any suitable volatile and/or non-volatile memory item (e.g., magnetic hard disk drive, solid state hard drive, semiconductor storage device, random access memory (RAM), read only memory (ROM), erasable programmable read only memory (EPROM), application specific integrated circuit (ASIC), etc.), software, logic (fixed logic, hardware logic, programmable logic, analog logic, digital logic), hardware, and/or in any other suitable component, device, element, and/or object as may be appropriate. Any of the memory items discussed herein should be construed as being encompassed within the broad term 'memory element'. Data/information being tracked and/or sent to one or more entities as discussed herein could be provided in any database, table, register, list, cache, storage, and/or storage structure: all of which can be referenced at any suitable timeframe. Any such storage options may also be included within the broad term 'memory element' as used herein.

(56) Note that in certain example implementations, operations as set forth herein may be implemented by logic encoded in one or more tangible media that is capable of storing instructions and/or digital information and may be inclusive of non-transitory tangible media and/or non-transitory computer readable storage media (e.g., embedded logic provided in: an ASIC, digital signal processing (DSP) instructions, software [potentially inclusive of object code and source code], etc.) for execution by one or more processor(s), and/or other similar machine, etc. Generally, memory element(s) **504** and/or storage **506** can store data, software, code, instructions (e.g., processor instructions), logic, parameters, combinations thereof, and/or the like used for operations described herein. This includes memory element(s) **504** and/or storage **506** being able to store data, software, code, instructions (e.g., processor instructions), logic, parameters, combinations thereof, or the like that are executed to carry out operations in accordance with teachings of the present disclosure.

(57) In some instances, software of the present embodiments may be available via a non-transitory computer useable medium (e.g., magnetic or optical mediums, magneto-optic mediums, CD-ROM, DVD, memory devices, etc.) of a stationary or portable program product apparatus, downloadable

file(s), file wrapper(s), object(s), package(s), container(s), and/or the like. In some instances, non-transitory computer readable storage media may also be removable. For example, a removable hard drive may be used for memory/storage in some implementations. Other examples may include optical and magnetic disks, thumb drives, and smart cards that can be inserted and/or otherwise connected to a computing device for transfer onto another computer readable storage medium.

(58) FIG. 6 illustrates a block diagram of a computing device **600** that may perform the functions of the meeting server(s) **110** described herein. The computing device **600** may include one or more processor(s) **602**, one or more memory element(s) **604**, storage **606**, a bus **608**, one or more network processor unit(s) **610** interconnected with one or more network input/output (I/O) interface(s) **612**, one or more I/O interface(s) **614**, and meeting server logic **620**. In various embodiments, instructions associated with the meeting server logic **620** is configured to perform the meeting server operations described herein, including those depicted by the flow charts for method **300** shown in FIG. 3 and method **400** shown in FIG. 4.

(59) In one form, a computer-implemented method is provided comprising: generating one or more animated files using an image of a user; storing the one or more animated files in a data store associated with the user; determining that the user is participating in an online meeting and the user is not transmitting video; obtaining the one or more animated files from the data store; and transmitting the one or more animated files for display playback on a loop on devices of participants participating in the online meeting with the user when the user is not transmitting video.

(60) In one example, the one or more animated files includes a plurality of emotion animated files and the method further comprises: obtaining a trigger corresponding to a particular emotion associated with the user; and transmitting, for display on the devices of the participants, a particular emotion animated file, of the plurality of emotion animated files, corresponding to the particular emotion. In another example, the plurality of emotion animated files includes one or more of a smiling file, a laughing file, and a nodding file. In another example, obtaining the trigger includes obtaining an indication of the particular emotion using face detection or gesture detection. In another example, obtaining the trigger includes obtaining an indication that the user has selected a reaction button.

(61) In another example, transmitting the particular emotion animated file comprises: obtaining a user identifier for the user associated with the particular emotion and an emotion identifier associated with the particular emotion animated file corresponding to the particular emotion; identifying the particular emotion animated file based on the emotion identifier and the user identifier; and transmitting the particular emotion animated file associated with the user identifier. In another example, the method further comprises transmitting the one or more animated files on the loop when the particular emotion animated file concludes. In another example, the image of the user includes a photographic image of the user and the one or more animated files include artificial intelligence models in which a head or facial features of the user in the photographic image move. In another example, the image of the user in the one or more animated files has a common expression at a beginning and a conclusion of the one or more animated files.

(62) In another form, an apparatus is provided comprising: a memory; a network interface configured to enable network communication; and a processor, wherein the processor is configured to perform operations comprising: generating one or more animated files using an image of a user; storing the one or more animated files in a data store associated with the user; determining that the user is participating in an online meeting and the user is not transmitting video; obtaining the one or more animated files from the data store; and transmitting the one or more animated files for display playback on a loop on devices of participants participating in the online meeting with the user when the user is not transmitting video.

(63) In yet another form, one or more non-transitory computer readable storage media is provided. The one or more non-transitory computer readable storage media is encoded with instructions that,

when executed by a processor of a user device, cause the processor to execute a method comprising: generating one or more animated files using an image of a user; storing the one or more animated files in a data store associated with the user; determining that the user is participating in an online meeting and the user is not transmitting video; obtaining the one or more animated files from the data store; and transmitting the one or more animated files for display playback on a loop on devices of participants participating in the online meeting with the user when the user is not transmitting video.

Variations and Implementations

(64) Embodiments described herein may include one or more networks, which can represent a series of points and/or network elements of interconnected communication paths for receiving and/or transmitting messages (e.g., packets of information) that propagate through the one or more networks. These network elements offer communicative interfaces that facilitate communications between the network elements. A network can include any number of hardware and/or software elements coupled to (and in communication with) each other through a communication medium. Such networks can include, but are not limited to, any local area network (LAN), virtual LAN (VLAN), wide area network (WAN) (e.g., the Internet), software defined WAN (SD-WAN), wireless local area (WLA) access network, wireless wide area (WWA) access network, metropolitan area network (MAN), Intranet, Extranet, virtual private network (VPN), Low Power Network (LPN), Low Power Wide Area Network (LPWAN), Machine to Machine (M2M) network, Internet of Things (IoT) network, Ethernet network/switching system, any other appropriate architecture and/or system that facilitates communications in a network environment, and/or any suitable combination thereof.

(65) Networks through which communications propagate can use any suitable technologies for communications including wireless communications (e.g., 4G/5G/nG, IEEE 802.11 (e.g., Wi-Fi®/Wi-Fi6®), IEEE 802.16 (e.g., Worldwide Interoperability for Microwave Access (WiMAX)), Radio-Frequency Identification (RFID), Near Field Communication (NFC), Bluetooth™ mm.wave, Ultra-Wideband (UWB), etc.), and/or wired communications (e.g., T1 lines, T3 lines, digital subscriber lines (DSL), Ethernet, Fibre Channel, etc.). Generally, any suitable means of communications may be used such as electric, sound, light, infrared, and/or radio to facilitate communications through one or more networks in accordance with embodiments herein. Communications, interactions, operations, etc. as discussed for various embodiments described herein may be performed among entities that may directly or indirectly connected utilizing any algorithms, communication protocols, interfaces, etc. (proprietary and/or non-proprietary) that allow for the exchange of data and/or information.

(66) Communications in a network environment can be referred to herein as ‘messages’, ‘messaging’, ‘signaling’, ‘data’, ‘content’, ‘objects’, ‘requests’, ‘queries’, ‘responses’, ‘replies’, etc. which may be inclusive of packets. As referred to herein and in the claims, the term ‘packet’ may be used in a generic sense to include packets, frames, segments, datagrams, and/or any other generic units that may be used to transmit communications in a network environment. Generally, a packet is a formatted unit of data that can contain control or routing information (e.g., source and destination address, source and destination port, etc.) and data, which is also sometimes referred to as a ‘payload’, ‘data payload’, and variations thereof. In some embodiments, control or routing information, management information, or the like can be included in packet fields, such as within header(s) and/or trailer(s) of packets. Internet Protocol (IP) addresses discussed herein and in the claims can include any IP version 4 (IPv4) and/or IP version 6 (IPv6) addresses.

(67) To the extent that embodiments presented herein relate to the storage of data, the embodiments may employ any number of any conventional or other databases, data stores or storage structures (e.g., files, databases, data structures, data or other repositories, etc.) to store information.

(68) Note that in this Specification, references to various features (e.g., elements, structures, nodes, modules, components, engines, logic, steps, operations, functions, characteristics, etc.) included in

‘one embodiment’, ‘example embodiment’, ‘an embodiment’, ‘another embodiment’, ‘certain embodiments’, ‘some embodiments’, ‘various embodiments’, ‘other embodiments’, ‘alternative embodiment’, and the like are intended to mean that any such features are included in one or more embodiments of the present disclosure, but may or may not necessarily be combined in the same embodiments. Note also that a module, engine, client, controller, function, logic or the like as used herein in this Specification, can be inclusive of an executable file comprising instructions that can be understood and processed on a server, computer, processor, machine, compute node, combinations thereof, or the like and may further include library modules loaded during execution, object files, system files, hardware logic, software logic, or any other executable modules.

(69) It is also noted that the operations and steps described with reference to the preceding figures illustrate only some of the possible scenarios that may be executed by one or more entities discussed herein. Some of these operations may be deleted or removed where appropriate, or these steps may be modified or changed considerably without departing from the scope of the presented concepts. In addition, the timing and sequence of these operations may be altered considerably and still achieve the results taught in this disclosure. The preceding operational flows have been offered for purposes of example and discussion. Substantial flexibility is provided by the embodiments in that any suitable arrangements, chronologies, configurations, and timing mechanisms may be provided without departing from the teachings of the discussed concepts.

(70) As used herein, unless expressly stated to the contrary, use of the phrase ‘at least one of’, ‘one or more of’, ‘and/or’, variations thereof, or the like are open-ended expressions that are both conjunctive and disjunctive in operation for any and all possible combination of the associated listed items. For example, each of the expressions ‘at least one of X, Y and Z’, ‘at least one of X, Y or Z’, ‘one or more of X, Y and Z’, ‘one or more of X, Y or Z’ and ‘X, Y and/or Z’ can mean any of the following: 1) X, but not Y and not Z; 2) Y, but not X and not Z; 3) Z, but not X and not Y; 4) X and Y, but not Z; 5) X and Z, but not Y; 6) Y and Z, but not X; or 7) X, Y, and Z.

(71) Additionally, unless expressly stated to the contrary, the terms ‘first’, ‘second’, ‘third’, etc., are intended to distinguish the particular nouns they modify (e.g., element, condition, node, module, activity, operation, etc.). Unless expressly stated to the contrary, the use of these terms is not intended to indicate any type of order, rank, importance, temporal sequence, or hierarchy of the modified noun. For example, ‘first X’ and ‘second X’ are intended to designate two ‘X’ elements that are not necessarily limited by any order, rank, importance, temporal sequence, or hierarchy of the two elements. Further as referred to herein, ‘at least one of’ and ‘one or more of’ can be represented using the ‘(s)’ nomenclature (e.g., one or more element(s)).

(72) Each example embodiment disclosed herein has been included to present one or more different features. However, all disclosed example embodiments are designed to work together as part of a single larger system or method. This disclosure explicitly envisions compound embodiments that combine multiple previously-discussed features in different example embodiments into a single system or method.

(73) One or more advantages described herein are not meant to suggest that any one of the embodiments described herein necessarily provides all of the described advantages or that all the embodiments of the present disclosure necessarily provide any one of the described advantages. Numerous other changes, substitutions, variations, alterations, and/or modifications may be ascertained to one skilled in the art and it is intended that the present disclosure encompass all such changes, substitutions, variations, alterations, and/or modifications as falling within the scope of the appended claims.

Claims

1. A computer-implemented method comprising: generating a plurality of animated files using an image of a user, the plurality of animated files being video representations of the user that are

created using the image of the user, the plurality of animated files including one or more basic animated files and one or more emotion animated files, each emotion animated file being associated with an emotion; storing the plurality of animated files in a data store associated with the user; determining that the user is participating in an online meeting and the user is not transmitting video; obtaining at least one basic animated file, of the one or more basic animated files, and at least one emotion animated file, of the one or more emotion animated files, from the data store; transmitting the at least one basic animated file for display playback on a loop on devices of participants participating in the online meeting with the user when the user is not transmitting video; obtaining a trigger corresponding to a particular emotion associated with the user; transmitting, for display playback on the devices of the participants, a particular emotion animated file, of the at least one emotion animated file, corresponding to the particular emotion, wherein the image of the user has a common expression at a beginning and a conclusion of the at least one basic animated file and the particular emotion animated file; and transmitting the at least one basic animated file for display playback on the loop when the particular emotion animated file concludes.

2. The computer-implemented method of claim 1, wherein the one or more emotion animated files includes one or more of a smiling file, a laughing file, and a nodding file.

3. The computer-implemented method of claim 1, wherein obtaining the trigger includes obtaining an indication of the particular emotion using face detection or gesture detection.

4. The computer-implemented method of claim 1, wherein obtaining the trigger includes obtaining an indication that the user has selected a reaction button.

5. The computer-implemented method of claim 1, wherein transmitting the particular emotion animated file comprises: obtaining a user identifier for the user associated with the particular emotion and an emotion identifier associated with the particular emotion animated file corresponding to the particular emotion; identifying the particular emotion animated file based on the emotion identifier and the user identifier; and transmitting the particular emotion animated file associated with the user identifier.

6. The computer-implemented method of claim 1, wherein the image of the user includes a photographic image of the user and the plurality of animated files include artificial intelligence models in which a head or facial features of the user in the photographic image move.

7. An apparatus comprising: a memory; a network interface configured to enable network communication; and a processor, wherein the processor is configured to perform operations comprising: generating a plurality of animated files using an image of a user, the plurality of animated files being video representations of the user that are created using the image of the user, the plurality of animated files including one or more basic animated files and one or more emotion animated files, each emotion animated file being associated with an emotion; storing the plurality of animated files in a data store associated with the user; determining that the user is participating in an online meeting and the user is not transmitting video; obtaining at least one basic animated file, of the one or more basic animated files, and at least one emotion animated file, of the one or more emotion plurality of animated files, from the data store; transmitting the at least one basic animated file for display playback on a loop on devices of participants participating in the online meeting with the user when the user is not transmitting video; obtaining a trigger corresponding to a particular emotion associated with the user; transmitting, for display playback on the devices of the participants, a particular emotion animated file, of the at least one emotion animated file, corresponding to the particular emotion, wherein the image of the user has a common expression at a beginning and a conclusion of the at least one basic animated file and the particular emotion animated file; and transmitting the at least one basic animated file for display playback on the loop when the particular emotion animated file concludes.

8. The apparatus of claim 7, wherein the processor is configured to perform the operation of obtaining the trigger by obtaining an indication of the particular emotion using face detection or gesture detection.

9. The apparatus of claim 7, wherein the processor is configured to perform the operation of obtaining the trigger by obtaining an indication that the user has selected a reaction button.
10. The apparatus of claim 7, wherein the processor is configured to perform the operation of transmitting the particular emotion animated file by: obtaining a user identifier for the user associated with the particular emotion and an emotion identifier associated with the particular emotion animated file corresponding to the particular emotion; identifying the particular emotion animated file based on the emotion identifier and the user identifier; and transmitting the particular emotion animated file associated with the user identifier.
11. The apparatus of claim 7, wherein the image of the user includes a photographic image of the user and the plurality of animated files include artificial intelligence models in which a head or facial features of the user in the photographic image move.
12. One or more non-transitory computer readable storage media encoded with instructions that, when executed by a processor of a user device, cause the processor to execute a method comprising: generating a plurality of animated files using an image of a user, the plurality of animated files being video representations of the user that are created using the image of the user, the plurality of animated files including one or more basic animated files and one or more emotion animated files, each emotion animated file being associated with an emotion; storing the plurality of animated files in a data store associated with the user; determining that the user is participating in an online meeting and the user is not transmitting video; obtaining at least one basic animated file, of the one or more basic animated files and at least one emotion animated file, of the one or more emotion animated files, from the data store; transmitting the at least one basic animated file for display playback on a loop on devices of participants participating in the online meeting with the user when the user is not transmitting video; obtaining a trigger corresponding to a particular emotion associated with the user; transmitting, for display playback on the devices of the participants, a particular emotion animated file, of the at least one emotion animated file, corresponding to the particular emotion, wherein the image of the user has a common expression at a beginning and a conclusion of the at least one basic animated file and the particular emotion animated file; and transmitting the at least one basic animated file for display playback on the loop when the particular emotion animated file concludes.
13. The one or more non-transitory computer readable storage media of claim 12, wherein obtaining the trigger includes obtaining an indication of the particular emotion using face detection or gesture detection.
14. The one or more non-transitory computer readable storage media of claim 12, wherein obtaining the trigger includes obtaining an indication that the user has selected a reaction button.
15. The one or more non-transitory computer readable storage media of claim 12, wherein transmitting the particular emotion animated file comprises: obtaining a user identifier for the user associated with the particular emotion and an emotion identifier associated with the particular emotion animated file corresponding to the particular emotion; identifying the particular emotion animated file based on the emotion identifier and the user identifier; and transmitting the particular emotion animated file associated with the user identifier.
16. The computer-implemented method of claim 1, further comprising determining that the user has selected an operating mode in which the one or more emotion animated files are to be used.
17. The apparatus of claim 7, wherein the one or more emotion animated files includes one or more of a smiling file, a laughing file, and a nodding file.
18. The apparatus of claim 7, wherein the processor is further configured to perform the operation of determining that the user has selected an operating mode in which the one or more emotion animated files are to be used.
19. The one or more non-transitory computer readable storage media of claim 12, wherein the one or more emotion animated files includes one or more of a smiling file, a laughing file, and a nodding file.

20. The one or more non-transitory computer readable storage media of claim 12, further comprising determining that the user has selected an operating mode in which the one or more emotion animated files are to be used.
